Topic 1: Acute Leukemia (AML & ALL)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Malignant clonal proliferation of immature hematopoietic blast cells replacing bone marrow. Acute Myeloid Leukemia (AML): adults (median age 65), Auer rods pathognomonic, myeloperoxidase positive. Acute Lymphoblastic Leukemia (ALL): children (peak age 2-5), TdT positive, mediastinal mass (T-cell ALL).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Bone marrow failure triad: 1) Anemia (fatigue, pallor, dyspnea); 2) Thrombocytopenia (petechiae, purpura, mucosal bleeding); 3) Neutropenia (frequent severe infections, fever). Bone pain, splenomegaly, leukostasis.
• **Physical Exam:** Pallor, diffuse petechiae, ecchymoses, gingival hyperplasia (AML M5 monocytic), lymphadenopathy and hepatosplenomegaly (prominent in ALL), fever.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC: variable WBC (leukopenia or extreme leukocytosis > 100,000/mcL with blast crisis), anemia, severe thrombocytopenia. Peripheral blood smear: > 20% blasts. AUER RODS = pathognomonic for AML. Bone marrow biopsy (gold standard). Flow cytometry, cytogenetics (t(15;17) in Acute Promyelocytic Leukemia [APL]).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Emergent Transfusions: Platelets if < 10,000/mcL or active bleeding. PRBCs for symptomatic anemia. Leukostasis Syndrome (WBC > 100k, dyspnea, stroke): Hydroxyurea, aggressive IV hydration, emergent leukapheresis. Tumor Lysis Syndrome prophylaxis (Allopurinol / Rasburicase + IV fluids). APL t(15;17) DIC emergency: STAT All-Trans Retinoic Acid (ATRA). Broad-spectrum antibiotics (Cefepime) for neutropenic fever (ANC < 500 + Temp >= 38.3°C).



HIGH-YIELD BOARD PEARL: Auer rods (crystalline cytoplasmic inclusions of fused lysosomes) are pathognomonic for Acute Myeloid Leukemia (AML). Acute Promyelocytic Leukemia (APL, t(15;17)) frequently presents with fulminant DIC and requires immediate administration of All-Trans Retinoic Acid (ATRA) prior to bone marrow biopsy confirmation.

Topic 2: Anemia Classification & Transfusion Triage

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Reduction in total red blood cell mass. Classified by Mean Corpuscular Volume (MCV): 1) Microcytic (MCV < 80 fL: Iron Deficiency, Thalassemia, Sideroblastic, Anemia of Chronic Disease); 2) Normocytic (MCV 80-100 fL: Acute Blood Loss, Hemolysis, Aplastic, Renal Failure); 3) Macrocytic (MCV > 100 fL: Megaloblastic [B12 & Folate deficiency] vs Non-megaloblastic [Alcohol, Liver disease, Hypothyroidism]).

CLINICAL PRESENTATION & PHYSICAL EXAM

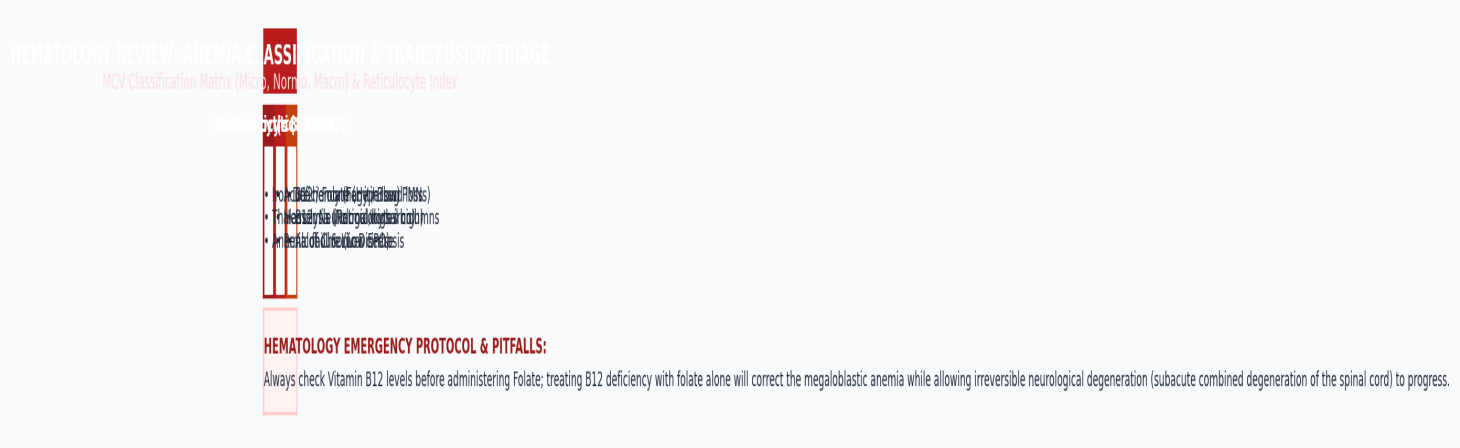
• **Presentation:** Fatigue, exertional dyspnea, weakness, lightheadedness, palpitations, syncope, angina in coronary patients.
• **Physical Exam:** Conjunctival and palmar pallor, resting tachycardia, systolic flow murmur. Koilonychia (spoon nails) in iron deficiency; glossitis and neurologic deficits (subacute combined degeneration / dorsal column loss) in B12 deficiency.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC with differential, Reticulocyte count (corrected reticulocyte index > 2% indicates hemolysis or active bleeding; < 2% indicates hypoproliferation). Ferritin, Iron, TIBC, Vitamin B12, Folate. Peripheral smear: Target cells (Thalassemia), Hypersegmented neutrophils (B12/Folate megaloblastic), Spherocytes (Hemolysis).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Emergency Transfusion Triggers: Restrictive transfusion threshold of Hemoglobin < 7 g/dL for stable hospitalized patients (target Hgb 7-9 g/dL). Threshold of Hemoglobin < 8 g/dL for acute coronary syndrome / cardiovascular disease. Active hemorrhagic shock: Transfuse regardless of initial hemoglobin! Target specific deficiency: Oral/IV Iron, B12 IM, Folate.



HIGH-YIELD BOARD PEARL: A restrictive transfusion strategy (Hemoglobin trigger < 7 g/dL) is standard of care and achieves equal or superior outcomes compared to liberal transfusion (trigger < 10 g/dL), except in acute myocardial ischemia where a target threshold of Hemoglobin >= 8 g/dL is recommended.

Topic 3: Aplastic Anemia

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Pancytopenia with severe bone marrow hypocellularity. Autoimmune T-cell mediated destruction of hematopoietic stem cells. Triggers: idiopathic (50%), drugs (Chloramphenicol, Carbamazepine, Sulfonamides, NSAIDs), toxic chemicals (Benzene), radiation, viral infections (Parvovirus B19, Hepatitis, EBV, HIV).

CLINICAL PRESENTATION & PHYSICAL EXAM

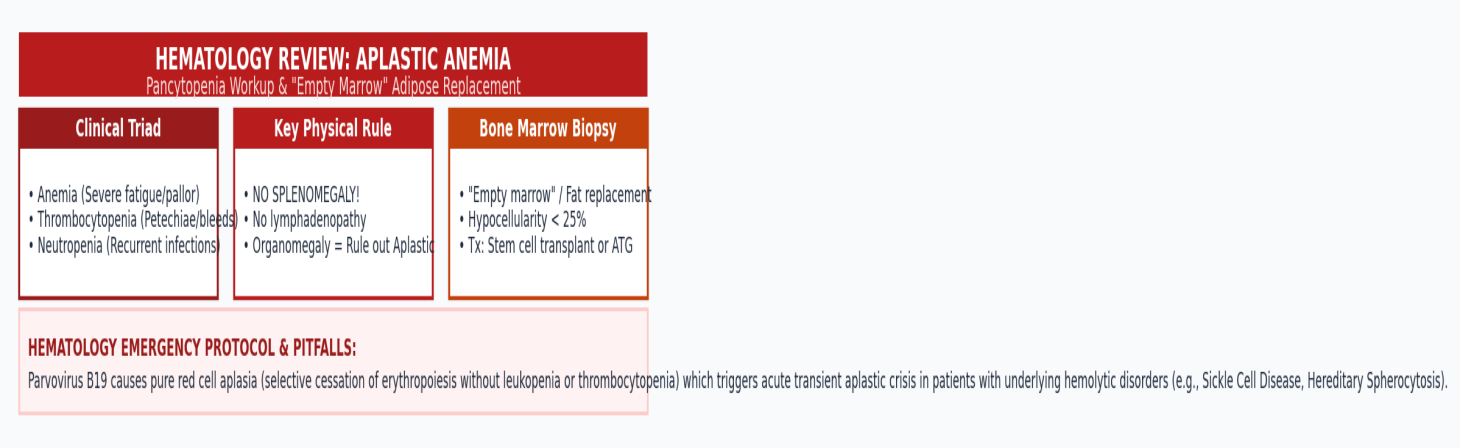
• **Presentation:** Insidious or acute onset of pancytopenia: severe fatigue and pallor (anemia), recurrent bacterial/fungal infections (neutropenia), and mucosal bleeding, purpura, petechiae, epistaxis (thrombocytopenia).
• **Physical Exam:** Marked pallor, ecchymoses, petechiae. ABSENCE of splenomegaly, hepatomegaly, or lymphadenopathy (presence of organomegaly points away from aplastic anemia toward leukemia or lymphoma).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC showing severe pancytopenia with absolute reticulocytopenia (reticulocyte count < 1%). Bone Marrow Biopsy (definitive gold standard): reveals profound hypocellularity with replacement of marrow by fat cells ("empty marrow").

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Strict reverse protective isolation (prevent infection). Transfusion support (leukoreduced, irradiated PRBCs and platelets to prevent HLA alloimmunization). Hematology consultation for definitive therapy: Allogeneic Hematopoietic Stem Cell Transplantation (HSCT) for young patients; Immunosuppressive therapy (Antithymocyte Globulin [ATG] + Cyclosporine) for older patients.



HIGH-YIELD BOARD PEARL: Bone marrow biopsy in aplastic anemia demonstrates a profound, dry, hypocellular marrow replaced by adipose tissue ("empty marrow"). Splenomegaly and lymphadenopathy are conspicuously ABSENT in aplastic anemia; their presence strongly suggests an alternative diagnosis like leukemia.

Topic 4: Clotting Factor Disorders (Hemophilia A/B & vWD)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

1) Hemophilia A: X-linked recessive deficiency of Factor VIII (80%). 2) Hemophilia B ("Christmas Disease"): X-linked recessive deficiency of Factor IX. 3) von Willebrand Disease (vWD): Autosomal dominant deficiency/dysfunction of vWF (most common inherited bleeding disorder; vWF bridges platelets to subendothelium and stabilizes Factor VIII).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Hemophilia A/B: Hemarthroses (spontaneous bleeding into weight-bearing joints: knees, elbows, ankles), deep intramuscular hematomas, prolonged bleeding post-trauma/surgery. vWD: Mucocutaneous bleeding (epistaxis, menorrhagia, gingival bleeding, easy bruising).
• **Physical Exam:** Tender, swollen, erythematous, warm, locked joint (acute hemarthrosis). Large palpable muscle hematomas (iliopsoas hematoma risks femoral nerve compression). Petechiae typical for platelet/vWD, not factor deficiency.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Hemophilia: PROLONGED aPTT with NORMAL PT/INR and NORMAL Platelet count. Mixing study: aPTT corrects with normal plasma (factor deficiency). Factor VIII/IX assays confirm diagnosis. vWD: Normal or prolonged aPTT, normal PT, decreased vWF antigen, decreased Ristocetin cofactor activity.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute Hemarthrosis / Bleed in Hemophilia A: STAT Factor VIII concentrate infusion. Target 100% factor activity for major/life-threatening bleeds (intracranial, retroperitoneal, compartment syndrome: 50 units/kg of Factor VIII; each unit/kg raises Factor VIII by 2%). Minor hemarthrosis: target 40-50% activity (20-25 units/kg). Hemophilia B: Factor IX concentrate (1 unit/kg raises Factor IX by 1%; give 100 units/kg for major bleed). Desmopressin (DDAVP 0.3 mcg/kg) for mild Hemophilia A and Type 1 vWD (releases endogenous vWF and Factor VIII from Weibel-Palade bodies). Antifibrinolytics (TXA).



HIGH-YIELD BOARD PEARL: Hemophilia A and B present with prolonged aPTT, normal PT/INR, and deep joint hemarthroses. In major or head-trauma bleeding in a hemophiliac, NEVER wait for lab results or CT confirmation—administer Factor replacement immediately to achieve 100% factor activity.

Topic 5: Microangiopathic Hemolytic Anemia (TTP, HUS, DIC)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Microvascular thrombosis causing mechanical shearing of RBCs. 1) Thrombotic Thrombocytopenic Purpura (TTP): ADAMTS13 metalloproteinase deficiency/inhibitor -> ultra-large vWF multimers shear RBCs and consume platelets. 2) Hemolytic Uremic Syndrome (HUS): Shiga toxin (E. coli O157:H7) endothelial damage in renal microvasculature. 3) DIC: Widespread systemic activation of coagulation cascade.

CLINICAL PRESENTATION & PHYSICAL EXAM

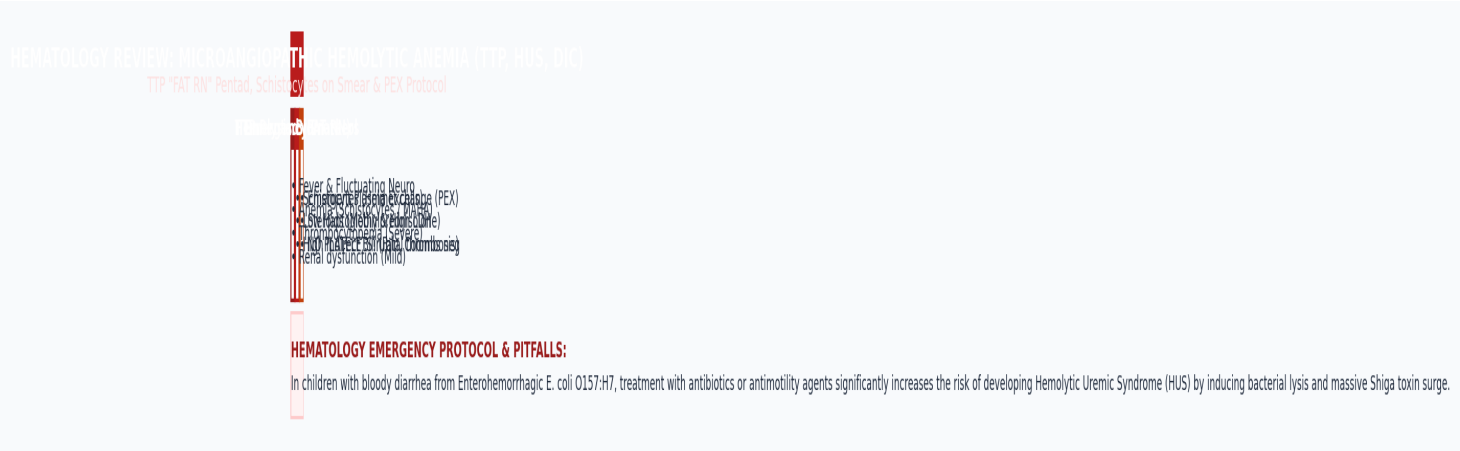
• **Presentation:** TTP Classic Pentad mnemonic "FAT RN": Fever, Anemia (microangiopathic hemolytic), Thrombocytopenia, Renal failure, Neurologic symptoms (confusion, seizures, fluctuating stroke-like deficits). HUS Triad: Anemia + Thrombocytopenia + Acute Renal Failure in a child following bloody diarrhea. DIC: Widespread bleeding from IV sites/orifices + thrombosis.
• **Physical Exam:** Jaundice, scleral icterus, petechiae, purpura, fluctuating mental status, oliguria.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Peripheral Blood Smear: SCHISTOCYTES (helmet cells, fragmented RBCs, > 2-3 per HPF confirms MAHA). Hemolysis lab profile: Elevated LDH, markedly elevated Indirect Bilirubin, DECREASED HAPTOGLOBIN (< 10 mg/dL), Reticulocytosis, negative direct Coombs test. Coagulation: Normal PT and aPTT in TTP and HUS (differentiates from DIC where PT, aPTT are prolonged and Fibrinogen is low!). Severely reduced ADAMTS13 activity (< 10%) confirms TTP.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

EMERGENT THERAPEUTIC PLASMA EXCHANGE (PEX / Plasmapheresis) is the definitive life-saving treatment for TTP (removes autoantibodies and replenishes ADAMTS13); initiate immediately. Glucocorticoids (Methylprednisolone 1 g IV) + Caplacizumab (anti-vWF nanobody). PLATELET TRANSFUSIONS ARE STRICTLY CONTRAINDICATED IN TTP (fuels microvascular thrombosis - "adding fuel to the fire" - leading to stroke, MI, and death). HUS: Supportive fluid care, avoid antibiotics for E. coli O157:H7 (increases Shiga toxin release). DIC: Treat underlying cause, Cryoprecipitate (if fibrinogen < 150), FFP, platelets.



HIGH-YIELD BOARD PEARL: The presence of Schistocytes on peripheral smear combined with thrombocytopenia is TTP until proven otherwise and mandates STAT Hematology consultation for emergent Plasma Exchange (PEX). Platelet transfusions are STRICTLY CONTRAINDICATED in TTP because they fuel microvascular thrombosis and precipitate fatal strokes and myocardial infarction.

Topic 6: Hypercoagulable States & Venous Thromboembolism (DVT)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Virchow Triad: Endothelial injury, Stasis of blood flow, and Hypercoagulability. Inherited: Factor V Leiden mutation (most common, resistance to activated protein C), Prothrombin G20210A mutation, Protein C/S deficiency, Antithrombin III deficiency. Acquired: Antiphospholipid Syndrome (Lupus anticoagulant, anticardiolipin), Malignancy (Trousseau sign of migratory thrombophlebitis), OCPs/Estrogen, Pregnancy, Nephrotic syndrome.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Deep Vein Thrombosis (DVT): Unilateral lower extremity swelling, dull aching pain, erythema, warmth. Pulmonary Embolism (PE): Sudden dyspnea, pleuritic chest pain, tachypnea, tachycardia.
• **Physical Exam:** Unilateral calf or thigh edema (calf circumference discrepancy > 3 cm measured 10 cm below tibial tuberosity). Palpable venous cord, local tenderness. Phlegmasia cerulea dolens: massive iliofemoral DVT causing painful blue cyanotic swollen leg and compartment syndrome.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Wells DVT Score risk stratification. Low risk: D-Dimer test (high sensitivity / negative predictive value to rule out). Moderate/High risk: Compression Venous Duplex Ultrasound (non-compressibility of vein confirms DVT).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Direct Oral Anticoagulants (DOACs: Apixaban 10 mg BID x 7d then 5 mg BID, or Rivaroxaban 15 mg BID x 21d then 20 mg daily) are first-line for non-cancer DVT. Low-Molecular-Weight Heparin (Enoxaparin 1 mg/kg SQ Q12H) preferred in pregnancy and active malignancy. Phlegmasia cerulea dolens: Emergent Catheter-Directed Thrombolysis (CDT) or surgical thrombectomy. IVC filter indicated ONLY if absolute contraindication to anticoagulation or recurrent PE despite therapeutic anticoagulation.



HIGH-YIELD BOARD PEARL: Factor V Leiden mutation is the most common inherited thrombophilia; the mutated Factor V is resistant to cleavage and inactivation by Activated Protein C. Direct Oral Anticoagulants (DOACs) are preferred over Warfarin for uncomplicated DVT due to rapid onset and no monitoring requirement.

Topic 7: Lymphomas: Hodgkin vs Non-Hodgkin

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Malignant neoplasms of lymphoid tissue. Hodgkin Lymphoma (HL): B-cell malignancy characterized by multinucleated giant REED-STERNBERG CELLS ("owl-eye" appearance), bimodal age distribution (20s and >65), strong association with Epstein-Barr Virus (EBV). Non-Hodgkin Lymphoma (NHL): diverse B-cell (85%) or T-cell neoplasms (Diffuse Large B-cell, Follicular, Burkitt lymphoma [t(8;14) c-myc, "starry sky" histology]), older adults.

CLINICAL PRESENTATION & PHYSICAL EXAM

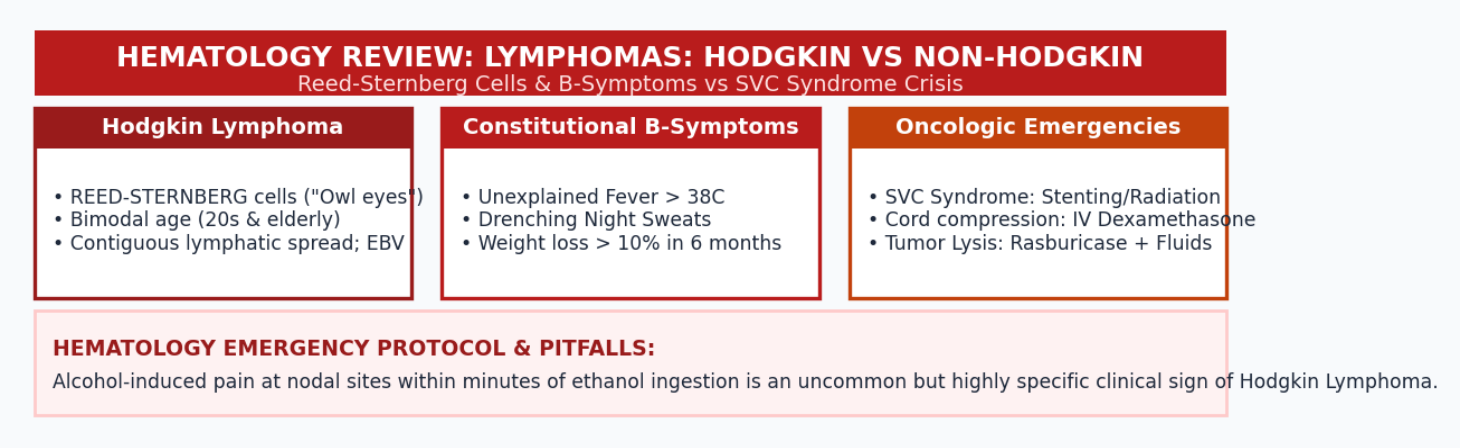
• **Presentation:** Painless, firm, rubbery lymphadenopathy (cervical, supraclavicular, axillary). Constitutional "B Symptoms" (indicates advanced stage / poor prognosis): 1) Fever (>38°C); 2) Drenching Night Sweats; 3) Unintentional Weight Loss (>10% over 6 months). Pel-Ebstein fevers (cyclical fevers recurring over 1-2 weeks in HL). Alcohol-induced lymph node pain in Hodgkin.
• **Physical Exam:** Non-tender, rubbery, mobile or matted lymphadenopathy. Superior Vena Cava (SVC) Syndrome: facial plethora, neck fullness, dilated chest wall collateral veins, arm swelling from mediastinal mass.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

EXCISIONAL LYMPH NODE BIOPSY is the gold standard (fine-needle aspiration [FNA] is INSUFFICIENT because architecture is required). Reed-Sternberg cells (CD15+, CD30+) confirm Hodgkin. CT Chest/Abdomen/Pelvis and PET-CT for Ann Arbor staging.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Hematology-Oncology referral for chemotherapy (ABVD regimen for Hodgkin; R-CHOP for DLBCL Non-Hodgkin) +/- radiation. Emergencies in ED: SVC Syndrome: Elevate head of bed, dexamethasone, emergent radiation or endovascular stenting. Spinal cord compression: High-dose Dexamethasone (10-20 mg IV) + STAT MRI spine. Tumor Lysis Syndrome: Aggressive IV hydration + Rasburicase.



HIGH-YIELD BOARD PEARL: Excisional lymph node biopsy is required for the definitive diagnosis of lymphoma; Fine Needle Aspiration (FNA) is inadequate because tissue architecture must be evaluated. Reed-Sternberg cells with prominent binucleated "owl-eye" nucleoli are pathognomonic for Hodgkin Lymphoma.

Topic 8: Polycythemia & Polycythemia Vera

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Elevated red blood cell mass (Hemoglobin > 16.5 g/dL in men, > 16.0 g/dL in women; Hematocrit > 49% men, > 48% women). Primary (Polycythemia Vera): myeloproliferative neoplasm driven by constitutively active JAK2 V617F mutation (>95% cases) with low erythropoietin. Secondary: appropriate hypoxia-driven (COPD, high altitude, cyanotic heart disease) or autonomous EPO production (Renal cell carcinoma, Hepatocellular carcinoma) with ELEVATED erythropoietin.

CLINICAL PRESENTATION & PHYSICAL EXAM

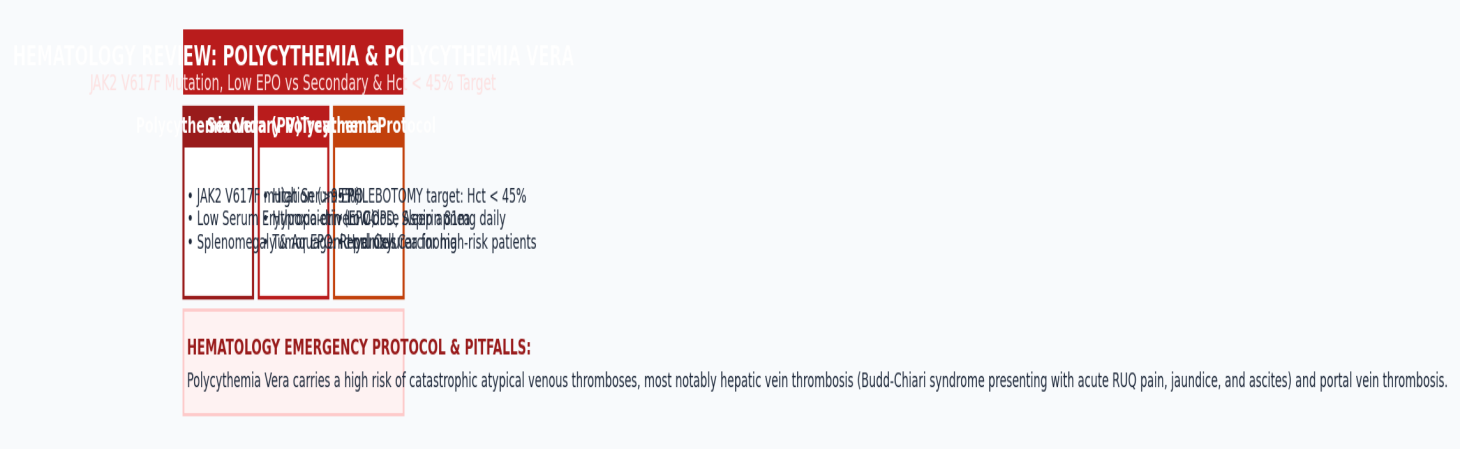
• **Presentation:** Hyperviscosity symptoms: headache, dizziness, visual disturbances, tinnitus, epistaxis. AQUAGENIC PRURITUS (severe intense itching after a warm shower or bath; hallmark of Polycythemia Vera). Erythromelalgia (burning pain and erythema in hands and feet). Venous or arterial thrombosis (Budd-Chiari syndrome, stroke, MI).
• **Physical Exam:** Plethoric "ruddy" facial complexion, engorged retinal veins on fundoscopy, splenomegaly (present in 75% of PV; absent in secondary polycythemia).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC: elevated RBC, Hgb, Hct, thrombocytosis, leukocytosis. Serum Erythropoietin (EPO) level (LOW in Polycythemia Vera; HIGH in secondary polycythemia). Peripheral blood JAK2 V617F mutation testing (positive in >95% PV). Bone marrow biopsy.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Therapeutic Phlebotomy is the cornerstone of emergency and long-term therapy: remove 500 mL of blood weekly until HEMATOCRIT IS MAINTAINED BELOW 45% (significantly reduces cardiovascular death and thrombosis). Low-dose Aspirin (81 mg daily) to prevent thrombotic complications. Cytoreductive therapy (Hydroxyurea) for high-risk patients (age > 60 or prior thrombosis).



HIGH-YIELD BOARD PEARL: Aquagenic pruritus (severe itching after a hot bath or shower) and a low serum erythropoietin level in the presence of elevated hematocrit are classic for Polycythemia Vera (>95% have JAK2 mutation). The therapeutic target is a Hematocrit maintained strictly below 45% via phlebotomy.

Topic 9: Sickle Cell Disease (Acute Crises & Resuscitation)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Autosomal recessive single nucleotide mutation in beta-globin gene (valine replaces glutamic acid at position 6; HbS). Under low oxygen, acidosis, or dehydration, HbS polymerizes into rigid sickle-shaped RBCs, causing vaso-occlusion, microvascular infarction, and severe chronic hemolytic anemia.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Acute Vaso-Occlusive Pain Crisis (VOC): excruciating pain in back, chest, extremities. Acute Chest Syndrome (ACS): new pulmonary infiltrate on CXR PLUS fever, chest pain, tachypnea, or hypoxia (leading cause of death!). Other crises: Aplastic crisis (Parvovirus B19), Splenic sequestration (infants with pooling of blood, shock, massive splenomegaly), Priapism (sustained ischemic erection > 4 hours).
• **Physical Exam:** Severe bony/joint tenderness, tachypnea, pallor, scleral icterus. Systolic murmur. Splenomegaly in young child (later autosplenectomy by age 5-6 leads to functional asplenia and encapsulated sepsis risk).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC (baseline Hgb 6-9 g/dL; drop > 2 g/dL indicates aplastic crisis or splenic sequestration), Reticulocyte count (elevated at baseline; RETICULOCYTOPENIA < 1% indicates Parvovirus B19 aplastic crisis). STAT Chest X-ray (mandatory in all patients with fever or chest pain to rule out ACS). Hemoglobin electrophoresis.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Vaso-occlusive Crisis: Rapid, aggressive intravenous OPIOID ANALGESIA (morphine or hydromorphone within 30-60 minutes of arrival; do not withhold opioids!), IV crystalloid hydration (avoid overhydration -> triggers pulmonary edema), supplemental oxygen ONLY if SpO2 < 95% (hyperoxia suppresses erythropoiesis). Acute Chest Syndrome: 1) Broad-spectrum IV antibiotics (Ceftriaxone + Azithromycin); 2) Simple or Exchange Transfusion (target HbS < 30% and Hgb ~10 g/dL); 3) Incentive spirometry Q2H. Priapism: Corporal aspiration and irrigation with Phenylephrine; exchange transfusion. Empiric IV Ceftriaxone for ALL febrile sickle cell patients (treat encapsulated organisms: Strep pneumoniae, Salmonella osteomyelitis).



HIGH-YIELD BOARD PEARL: Acute Chest Syndrome is the leading cause of mortality in Sickle Cell Disease; it is defined as a new pulmonary infiltrate on chest X-ray accompanied by fever, chest pain, or respiratory distress, and is treated with exchange transfusion, broad-spectrum antibiotics, and incentive spirometry.

Topic 10: Thrombocytopenia (ITP, TTP, HIT & Resuscitation)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Platelet count < 150,000/mcL. 1) Immune Thrombocytopenic Purpura (ITP): Anti-GpIIb/IIIa autoantibodies causing splenic macrophage destruction of platelets. 2) Heparin-Induced Thrombocytopenia (HIT): IgG antibodies against Heparin-Platelet Factor 4 (PF4) complex triggering paradoxically catastrophic THROMBOSIS ("white clot syndrome"). 3) TTP: ADAMTS13 deficiency. 4) Gestational thrombocytopenia.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Mucocutaneous bleeding: petechiae, purpura, epistaxis, gingival bleeding, menorrhagia. Life-threatening: intracranial hemorrhage (headache, altered mental status), massive GI bleeding. HIT: DVT, pulmonary embolism, skin necrosis at heparin injection sites 5-10 days after heparin initiation.

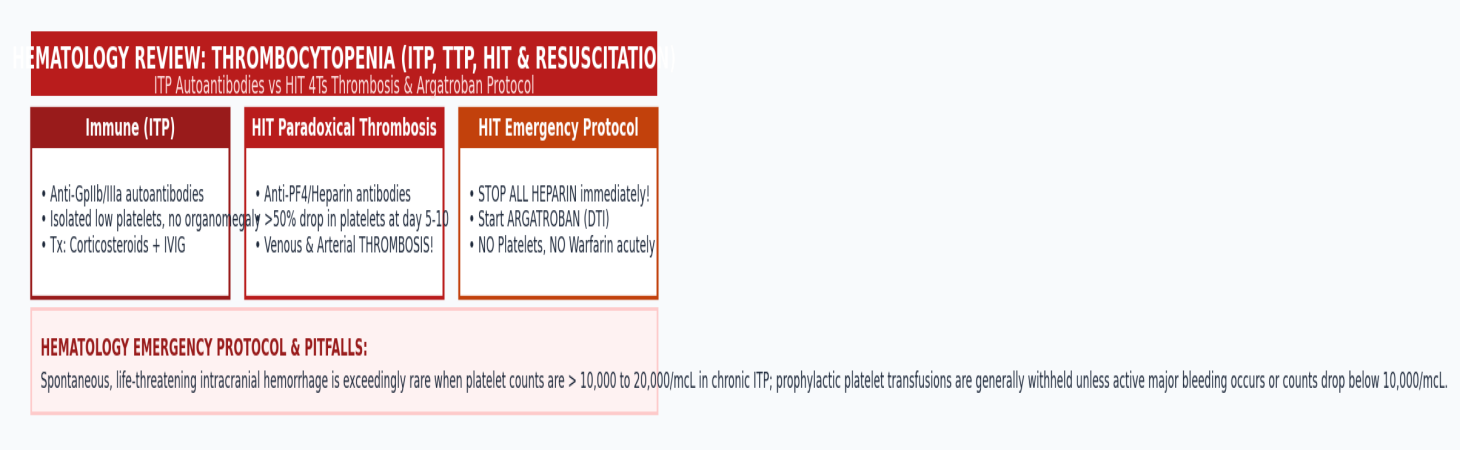
• **Physical Exam:** Diffuse petechiae (non-blanching macules < 3 mm) on dependent extremities, wet purpura in oral mucosa (heralds imminent intracranial hemorrhage). Absence of hepatosplenomegaly in isolated ITP.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CBC (isolated thrombocytopenia in ITP, smear shows large platelets/megakaryocytes). HIT: 4Ts Score (Thrombocytopenia >50% drop, Timing 5-10 days, Thrombosis, oTher causes ruled out); ELISA for anti-PF4-heparin antibodies, Serotonin Release Assay (gold standard). Check PT/INR and PTT (normal in ITP/TTP/HIT; prolonged in DIC).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

ITP Treatment: Indicated if Platelets < 30,000/mcL or active bleeding: First-line is High-dose Dexamethasone (40 mg daily x 4 days) OR Prednisone 1 mg/kg/day + IVIG (1 g/kg). Emergent life-threatening bleed (ICH): STAT Platelet Transfusion + IVIG 1 g/kg + Methylprednisolone 1 g IV. HIT Management: IMMEDIATELY DISCONTINUE ALL HEPARIN (including flushes/coated catheters). START DIRECT THROMBIN INHIBITOR (Argatroban or Bivalirudin) immediately. DO NOT GIVE WARFARIN (triggers venous limb gangrene; wait until platelets recover > 150k). DO NOT GIVE PLATELETS IN HIT.



HIGH-YIELD BOARD PEARL: In Heparin-Induced Thrombocytopenia (HIT), a 50% drop in platelet count typically occurs 5 to 10 days after heparin exposure and is paradoxical because it causes ARTERIAL AND VENOUS THROMBOSIS, not bleeding. Immediately discontinue all heparin and initiate a Direct Thrombin Inhibitor like Argatroban; platelet transfusions are strictly contraindicated.